BUSINESS INSIGHTS FROM ONLINE RETAIL DATA

TURNING DATA INTO DECISIONS

Data | Insights | Strategy

OBJECTIVES & PROBLEM STATEMENT

Problem Statement:

Online retailers face challenges in retaining customers beyond their first purchase and sustaining revenue growth.

Our Goals:

- Analyze customer retention and revenue trends using the Online Retail dataset (2009–2011).
- Identify key performance indicators (KPIs) to measure customer behavior and business performance.
- Perform deep-dive cohort analysis to uncover retention patterns.
- Build an interactive dashboard to visualize insights.
- Validate findings statistically to ensure business confidence.

Dataset Overview: Source & Composition

Source: Online Retail dataset (Kaggle)

Size: Approximately 500,000 transactional records.

Scope: Covers customer purchases across multiple countries.

Key Variables:

Variable Name	Description	Data Type
InvoiceNo	Unique identifier for each transaction	String
CustomerID	Unique identifier for each customer	Integer
Quantity	Number of items purchased per transaction	Integer
UnitPrice	Price per item	Float
InvoiceDate	Date and time of transaction	DateTime
Country	Customer's location	String

KEY FINDINGS FROM EXPLORATORY DATA ANALYSIS

- **Retention Trends:** Customer retention drops sharply after Month 1, stabilizing around 25–30% by Month 3.
- **Revenue Growth:** Revenue in 2010–2011 increased despite fewer orders.
- **Purchasing Behavior:** Average Order Value decreased slightly → customers shifted to smaller but more frequent purchases.
- **Seasonal Cohorts:** December 2009 cohort showed stronger retention compared to other months.
- **Geographic Insights:** UK contributed the largest share of customers and revenue.

VISUAL INSIGHTS FROM DATA ANALYSIS

Retention

Customer retention falls sharply after the first month, but then stabilizes around 25–30% by Month 3. This shows the importance of engaging customers early to keep them active.

Revenue Trend

Revenue grew from ~$9.54M in 2009–2010 to ~$9.75M in 2010–2011, even though the number of orders declined. This indicates stronger sales efficiency.

KPIs

2010–2011 saw ~500K customers and ~370K orders. Revenue reached ~$9.75M, slightly higher than the previous year. The average order value was ~$28, reflecting smaller but more frequent purchases.

Geography

The UK contributed nearly 70% of total revenue, making it the dominant market. Other key contributors were the Netherlands, Germany, France, and Ireland.

STRATEGIC BUSINESS INSIGHTS

Opportunities:

Retention Focus: Early engagement programs can reduce Month 1 churn and boost long-term value.

Revenue Growth: Higher revenue despite fewer orders shows potential in pricing and product mix.

UK Market Strength: With ~70% revenue from the UK, loyalty and cross-selling can be expanded.

International Expansion: Markets like NL, DE, FR, and IE offer scope for targeted growth.

Challenges:

Lower AOV: Average Order Value fell to ~$28, showing customers are buying smaller baskets.

Order Decline: Fewer orders in 2010–2011 may indicate reduced purchase frequency or market saturation.

Retention Drop: Sharp fall after Month 1 highlights high churn risk.

Geographic Dependence: Heavy reliance on the UK market increases vulnerability to regional shifts.

RECOMMENDATIONS

- **Boost Retention:** Launch onboarding campaigns and personalized offers to engage customers in the first month.
- **Increase AOV:** Introduce product bundles, upselling, and loyalty discounts to grow basket size.
- **Optimize Product Mix:** Focus on high-margin items and refine pricing strategies to sustain revenue growth.
- **Strengthen UK Market:** Expand loyalty programs and cross-selling initiatives in the core market.
- **Expand Internationally:** Target Netherlands, Germany, France, and Ireland with localized campaigns.
- **Diversify Markets:** Reduce dependence on the UK by investing in emerging regions.

CONCLUSION

Key Takeaway: Analysis shows strong revenue growth but challenges in retention, order volume, and market dependence.

Strategic Focus: Improve early engagement, increase basket size, and diversify beyond the UK.

Call to Action: Implement targeted retention programs, optimize product mix, and expand internationally for sustainable growth.